Reasoning, In Plain Sight

HMMT Problem Let ABC be a triangle with $\angle BAC = 90°$. Points X and Y are such that B, X, Y, and C lie on segment $\overline{BC}$ in that order, BX = 4, XY = 5, and YC = 3. Let T be a point lying on segment $\overline{AC}$ such that TA = TX = TY = ℓ for some ℓ. Compute ℓ.

Extracted Hidden CoT

Text of our custom "forced-reasoning" tool.

GPT-6 Astra

[...] Let d=CT=sqrt(h²+121/4) =sqrt(ell²+24). [...] TA=s-d=ell (T between) ->66/d=d+ell -> d ell= 42-ell². [...] Can simpler power T circle: TA TC = TX? power circle BT? [...] Then CT=CA-ell, relation. coordinate concise. [...]

GPT-5.6 Sol

[...] Because T on line AC and TA=TC? Not TC unknown. [...] Let z=h^2. Equation (z+25/4)(z+121/4)= (143/4-z)^2. Solve. [...] Need ensure existence and ordering T on AC, A proper. [...] Need maybe there are T below too symmetric. [...] T is A + l u yes. Perfect. Concise solution and exact required ending. [...]

Claude Opus 4.8

[...] Wait, T is on AC with TA=TX=TY. Let me set up. [...] Let me put A at origin, with AB and AC along axes. Actually let's use BC on x-axis. [...] So AC² = 12(12-a). Good. [...] Let me verify: 6.25 = 25/4, 30.25=121/4. [...] TX²=25/4+ (121/4)/3 = 25/4 + 121/12 = 75/12+121/12=196/12=49/3. Yes. [...]

Claude Sonnet 5

[...] wait is that right? [...] Wait that's not right, let's redo: [...] So d^2 = 12(12-a), nice. [...] matches! Great, so ℓ = 7√3/3. [...] Let's double check TX: TX^2=(6.5-4)^2+h^2 = 2.5^2 + 10.0833 = 6.25+10.0833=16.333 ✓. [...] Great, consistent. Also check TA: T=(6.5, 11√3/6), A=(3,3√3). [...] All consistent. So answer ℓ = 7√3/3. [...]

Answer Text

Text from the model's final response.

GPT-6 Astra

[...] so $108\ell^2=1764$. Since $\ell>0$, we obtain $\ell=7\sqrt{3}/3$.

FINAL_ANSWER: $7\sqrt{3}/3$

GPT-5.6 Sol

[...] Therefore $\ell^2=121/12 + 25/4 = 49/3$, so $\ell=7\sqrt{3}/3$.

FINAL_ANSWER: $7\sqrt{3}/3$

Claude Opus 4.8

[...] Then $\ell^2 = TX^2 = 25/4 + 121/12 = 49/3$, so $\ell = 7\sqrt{3}/3$.

FINAL_ANSWER: $7\sqrt{3}/3$

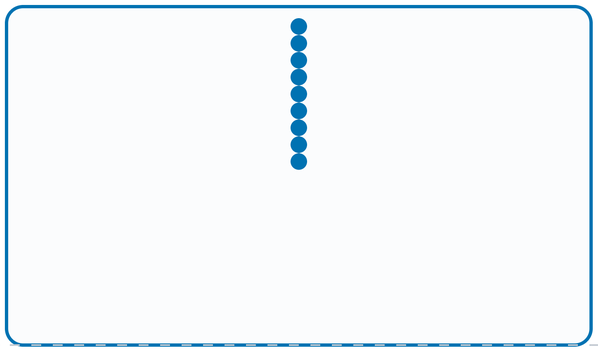
Claude Sonnet 5

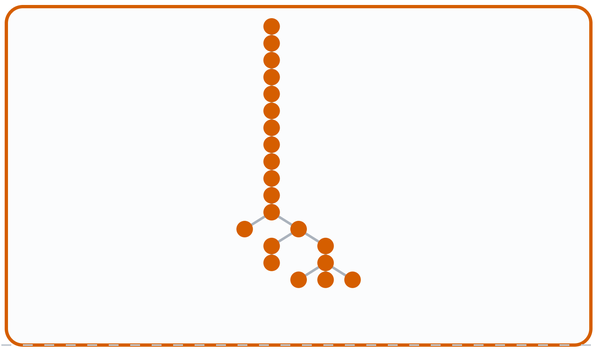
[...] $\ell = (60-18)/(6\sqrt{3}) = 42/(6\sqrt{3}) = 7/\sqrt{3} = 7\sqrt{3}/3$. [...]

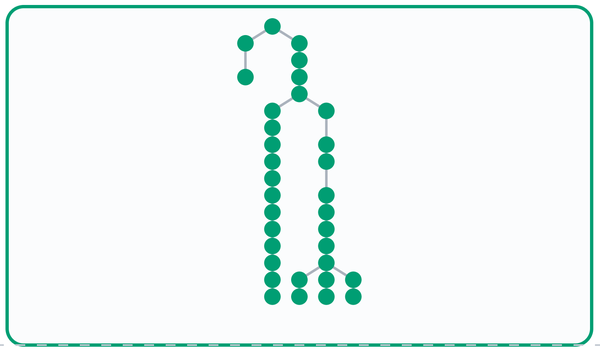
FINAL_ANSWER: $7\sqrt{3}/3$

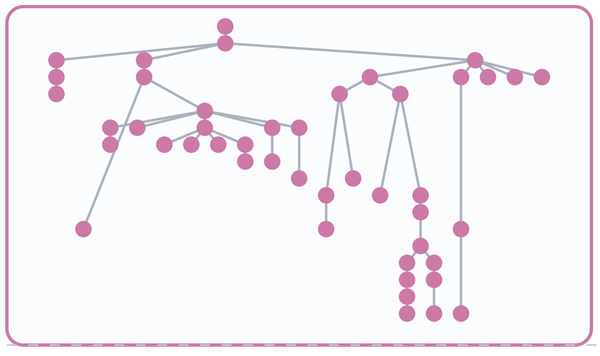
CoT Structure Tree

Nodes represent reasoning steps; branches indicate revisiting or pursuing another path.









Length & Redundancy

zlib compressed/original bytes · a lower ratio indicates more compressible text

GPT-6 Astra	GPT-5.6 Sol	Claude Opus 4.8	Claude Sonnet 5
757 chars & 0.622 zlib ratio	**1,932 chars & 0.531 zlib ratio**	**2,522 chars & 0.470 zlib ratio**	**11,455 chars & 0.375 zlib ratio**